PGRR: Planning-Guided Failure-Triggered Recovery and Rejoin for Dynamic Social Navigation

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Abstract—**Classical navigation is reliable in routine mapped environments yet remains brittle under reciprocal pedestrian interactions. We present PGRR, Planning-Guided Failure-Triggered Recovery and Rejoin, a layer that preserves Nav2 DWB for nominal navigation and invokes a learned policy only when observable risk or progress rules persist. Rather than commanding velocity, the policy selects among 21 temporary subgoals and four interpretable recovery modes. A planning mask rejects infeasible decisions, and an independent stopping-distance supervisor may override motion. A privileged short-horizon planner supplies recovery demonstrations for behavior cloning and a two-round DAgger workflow; checkpoint selection uses validation only. Deployment uses only LiDAR and navigation history. The held-out, three-density Gazebo benchmark evaluates 5 methods on the same 120 conditions each (600 method–episodes). PGRR reaches 90.8% of goals versus 70.8% for DWB, with paired PGRR-minus-DWB goal, collision, and timeout differences of** +20.0 [+12.5, +28.3] pp**,** −29.2 [−37.5, −21.7] pp**, and** +1.7 [+0.0, +4.2] pp**, respectively. Reporting all terminal outcomes separately prevents reduced contact from being conflated with task completion.**

I. Introduction

Classical navigation stacks remain attractive because mapping, global planning, local collision avoidance, and safety checks are explicit and independently testable. Their local planners can nevertheless freeze, oscillate, or enter reciprocal deadlock when moving pedestrians block a narrow passage. End-to-end learning can encode interaction patterns, but replacing the nominal controller also changes behavior in routine states where the classical stack is already competent.

This work asks whether learning can instead be confined to the short interval in which a classical planner is at risk of failure. PGRR, Planning-Guided Failure-Triggered Recovery and Rejoin, operates as a layer and leaves Nav2 DWB in control during normal PointGoal navigation. An observable detector and a hysteretic state machine invoke recovery only for collision risk, freeze, oscillation, deadlock, or explicit planner failure. The policy never commands wheel velocity: it selects one of 21 local temporary goals or WAIT, BACKUP, REPLAN, and CONTINUE. Nav2 executes the decision, after which PGRR restores the original task goal. A planning-derived action mask restricts selectable actions, and an independent braking supervisor constrains executed motion.

Training exploits simulator state without leaking it to deployment. A short-horizon expert rolls out every legal recovery action using the map and privileged pedestrian state. Behavior cloning initializes a compact policy, and DAgger relabels states visited by the learned controller. At test time, PGRR receives only LiDAR history, the local path and goal, robot and planner commands, progress history, planner status, and rule-detector scores.

The contributions are:

- a failure-triggered ROS2 hierarchy that preserves classical planning authority outside explicitly detected recovery intervals;
- a fixed, interpretable recovery action space with planning and sensor masks enforced during expert labeling and online policy execution;
- an automatically labeled BC–DAgger pipeline that aggregates learner-induced recovery states, with scenario-disjoint offline diagnostics of aggregation and mask dependence; and
- a predeclared paired five-method Gazebo benchmark over eight pedestrian-interaction families, three densities, five held-out repetitions per cell, immutable raw streams, and complete-condition statistical analysis.

The evaluation reports contact, task completion, timeout, planner failure, and intervention behavior separately. Its numerical claims are generated from the same five-method result table used for confidence intervals and paired tests; the fail-closed paper build accepts no incomplete or exploratory result stream. For historical context only, the earlier v1 frozen test recorded 0/24 PGRR collisions versus 19/24 for Base, but 16/24 PGRR timeouts versus 0/24 and only 8/24 versus 5/24 goal reaches; the adjusted goal-reaching comparison was not significant. This safety–completion trade-off is not moderate-v6 evidence and is never pooled with the new benchmark.

II. Related Work

A. Classical and Social Navigation

The dynamic-window approach searches dynamically feasible velocity commands over a short horizon [1]; Nav2 provides a modern ROS2 navigation system in which DWB implements this family of local control [2]. Reciprocal methods such as ORCA [3] and learned crowd-aware policies such as SARL [4] instead model interaction more directly. The broader social-navigation literature emphasizes that collision avoidance, comfort, legibility, and task efficiency are distinct objectives [5]. PGRR retains DWB as the nominal controller and evaluates geometric clearance and discomfort time; it does not claim to model all social norms.

B. Hybrid Planning and Recovery

All-in-One learns a planner switch that remains active throughout navigation [6], whereas DR-MPC learns a residual

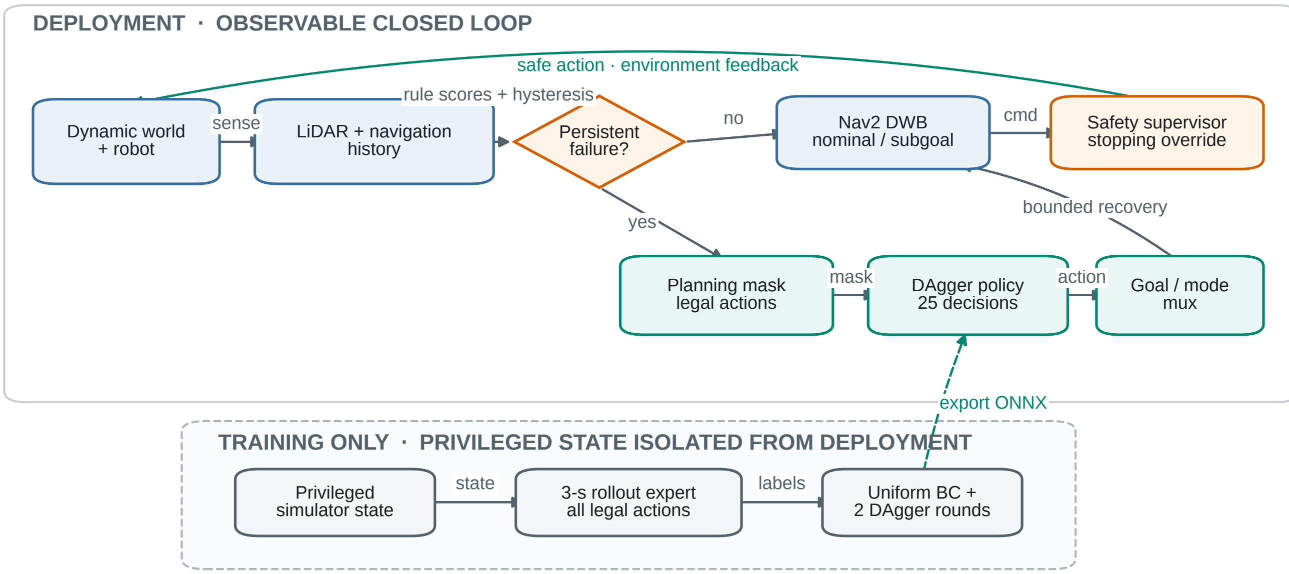

Fig. 1. Observable closed-loop PGRR architecture. Privileged simulator state and expert rollouts are confined to training (dashed region); deployment closes the sensing–planning–execution loop with LiDAR and navigation history while Nav2 retains nominal and temporary-goal execution.

around model-predictive control [7]. Closely related waypoint-generator systems insert an intermediate planner to guide a DRL obstacle-avoidance controller from a conventional global plan [8]. These works demonstrate the value of hybrid model-based and learned navigation. PGRR differs in authority and output: DWB executes both nominal and temporary goals, while learning is invoked only to select a bounded recovery option after persistent failure evidence.

Navigation recovery itself is not new. Del Duchetto *et al.* learn local failure detection and recovery from human demonstrations [9]; Mohammad *et al.* predict planning failure and search for a recovery state using a Gaussian-process model [10]. Recent preprints address failure-aware learned navigation from different angles: Fare embeds OOD detection and recognition in visual imitation policies and pairs them with heuristic recovery [11], while Wang *et al.* use failure experience to shape value estimation in offline navigation learning [12]. Our focus is automatic privileged-planner labels, planning-mask enforcement, and aggregation on states induced by the recovery learner. DAgger supplies the standard reduction for this sequential distribution shift [13]. Invalid-action masking has also been analyzed in policy-gradient learning [14]; here it is used as a deterministic planning constraint for imitation inference, not as a safety proof.

C. Simulation and Evaluation

Arena-Bench provides reproducible dynamic-obstacle evaluation [15], and Arena-Rosnav 2.0 adds unified interfaces for planners, simulators, and evaluation [16]. Later Arena 5.0 expands photorealistic social-navigation support [17]. Our experiments use a pinned Humble/Gazebo backend and a deterministic pedestrian protocol; that later release provides platform context rather than the experimental backend.

III. Problem Formulation and Method

A. Problem and Observable State

Let a differential-drive robot navigate a known static occupancy map from pose x_0 to goal g while a classical planner B supplies command u_t^B. PGRR must recover without pedestrian truth at deployment. Its history o_t contains five 180-beam LiDAR scans, goal distance and bearing, eight robot-frame path waypoints, robot velocity, u_t^B, ten-step goal-distance and angular-velocity histories, planner status, and four rule scores. Pedestrian truth is isolated for expert labeling and evaluation, as shown in Figure 1.

B. Failure Trigger and Recovery State Machine

Observable rules score collision risk, freeze, oscillation, and deadlock; Nav2 planner failure is an explicit status-triggered branch. Their aggregate is deliberately transparent:

$$s_t = \max\{s_t^{\text{col}}, s_t^{\text{frz}}, s_t^{\text{osc}}, s_t^{\text{dlk}}\} \in [0, 1]. \tag{1}$$

The temporal predicates are binary; collision risk may take an intermediate value from TTC or a bearing-consistent closing trend. Collision risk combines braking clearance, 1.5-s time to collision, near-field range, and a 0.5-s ego-compensated closing trend. Freeze requires more than 1 m remaining, less than 0.15 m displacement over 3 s, and a planner motion request. A Nav2 ABORTED/no-valid-control status supplies that request for the planner-failure branch; after the temporal gate confirms no motion, the recovery selector prioritizes a legal path subgoal and then REPLAN. Oscillation requires six angular-velocity sign changes beyond a 0.08 rad/s deadband and less than 0.20 m progress over 4 s. Deadlock combines low speed, an obstacle within 1.2 m, and less than 0.15 m displacement and goal progress over 4 s. Motion-history rules begin after a 4-s startup guard; collision evidence remains active.

Figure 2 shows the bounded state machine. NORMAL, PENDING, RECOVERY, and REJOIN are nonterminal; EMERGENCY-STOP has highest priority. Entry requires one decision frame above $\tau_{on} = 0.65$; exit requires four frames below $\tau_{off} = 0.35$ and renewed goal progress. Decisions run at 2 Hz with a 0.5-s hold and 2-s cooldown. Ordinary actions, directional yielding, the full recovery–rejoin sequence, and each REJOIN are bounded by 8, 30, 45, and 5 s, respectively; the sequence allows two returns to RECOVERY and four activations. PGRR stores the original goal before intervention and reissues it on REJOIN. Recovery success requires a REJOIN-to-NORMAL transition with that goal active, low failure score, and valid progress; final-goal reaching remains a separate endpoint.

C. Recovery Actions, Mask, and Supervisor

The 25 actions are 21 robot-relative subgoals

$$(r, \theta) \in \{0.6, 1.0, 1.4\}\,\mathrm{m} \times \{-90, -60, -30, 0, 30, 60, 90\}^\circ, \quad (2)$$

plus WAIT, BACKUP, REPLAN, and CONTINUE. The Boolean mask $M(o_t, a)$ rejects off-map or occupied endpoints, disconnected free space, obstructed swept segments, insufficient rear observability, excessive path deviation, and unavailable replanning. Specifically, the map segment must have 0.25-m clearance and its endpoint must be connected to the robot within a 10,000-cell search. The deployable LiDAR mask requires a 0.48-m swept capsule and endpoint clearance $r + 0.25$ m; BACKUP requires an observed 0.45-m rear segment plus 0.25 m clearance. The ordinary path envelope is 0.65 m; repeated learned recovery or observable emergency escalation widens that bounded regularizer to 2.5 m, without relaxing the other mask intersections. Collision latching raises motion clearance to 0.90 m. Pedestrian identity is absent online: dynamic occupancy enters through LiDAR, while privileged human positions may only further intersect the training mask. WAIT is the fail-closed fallback; mask-invalid logits are suppressed before softmax.

Path-frame scan differencing identifies persistent unilateral closing flow. A hysteretic yield latch and route-consistent side commitment then remove only already-legal actions directed toward the observed conflict. Repeated activation escalates to legal lateral subgoals or REPLAN, with a bounded BACKUP pulse when no lateral escape remains. These validation-selected filters only intersect the planning mask: they never re-enable motion and carry no isolated causal or formal-safety claim.

Emergency escape retains control through bounded retreat and turn pulses. The forward and reverse sectors meet at 80°, closing the directional gap without authorizing a translation that fails the swept-footprint mask. A turn requires 0.60 m of drift-aware LiDAR clearance; otherwise the controller keeps the rear-gated retreat option. Escape budgets reset only after at least 3 s of continuous clearance together with 0.25 m net progress toward the stored original goal. Thus a short release followed by motion into the same local minimum cannot renew the complete BACKUP/TURN budget.

An independent supervisor may override every action using

$$d_{stop} = \frac{v^2}{2a_b} + v\ell + m, \quad (3)$$

where $a_b = 0.8$ m/s^2, $\ell = 0.15$ s, and $m = 0.45$ m. It logs bounded stop and escape modes separately from policy output and applies footprint-aware translation and rotation clearances. It is an empirical filter, not a formal collision-avoidance guarantee.

D. Privileged Planning Expert

At training time, the expert combines LiDAR-derived local occupancy with privileged robot and pedestrian positions; velocities are finite-difference estimates rather than future simulator truth. Constant-velocity prediction with uncertainty inflation supports 3-s, 0.1-s rollouts. Subgoals use grid A* and pure pursuit with differential-drive kinematics, while the four modes use matched short-horizon models. The cost is

$$J(a) = w_c J_c + w_p J_p + w_s J_s + w_r J_r + w_l J_l \quad (4)$$

$$+ w_{sm} J_{sm} + w_t J_t + w_{sw} J_{sw} + w_w J_w, \quad (5)$$

for collision, progress, social distance, rejoin distance, length, smoothness, time, side switching, and repeated waiting. The weights are $(10^6, 8, 4, 3, 0.4, 0.2, 0.3, 0.5, 1)$; invalid actions are removed and collision receives a dominating finite penalty. The label stores the best action, all 25 costs, and the best-to-second-best margin. All candidates share the same 3-s horizon; no train/test-dependent cost normalization is used. When a scale-free gap is needed for an offline diagnostic, it is

$$\bar{\Delta} J(a) = \frac{[J(a) - J(a^*)]_+}{1 + |J(a^*)|}, \qquad a^* = \arg\min_{a: M(a)=1} J(a), \quad (6)$$

which leaves the expert action unchanged and is not an additional loss in the selected uniform-BC/DAgger model. Figure 2 illustrates this training-only ranking.

E. Behavior Cloning and DAgger

Two 1-D convolutions encode the 5×180 LiDAR stack; a 128-unit state encoder and two-layer head produce 25 logits (59,193 parameters). Uniform behavior cloning minimizes

$$\mathcal{L}_{BC} = -\frac{1}{N} \sum_i \log \pi_\theta(a_i^* \mid o_i, M_i). \quad (7)$$

DAgger executes the learner only on training scenarios, relabels visited recovery states with the expert, and retrains on the aggregate. Both planned rounds were run, but validation did not select the second-round candidate. The deployed checkpoint retrains the accepted aggregate with a train-only head-on coverage shard; validation and test episodes never enter it. It contains 1,387 raw states from 12 episodes, mirror-augmented to 2,774 samples, with 399 scenario-disjoint validation states. AdamW uses learning rate 3×10^{-4}, batch size 128, and early stopping. Selection uses validation loss and closed-loop validation only. PGRR denotes this uniform-BC/DAgger policy; PPO is not part of the evaluated method.

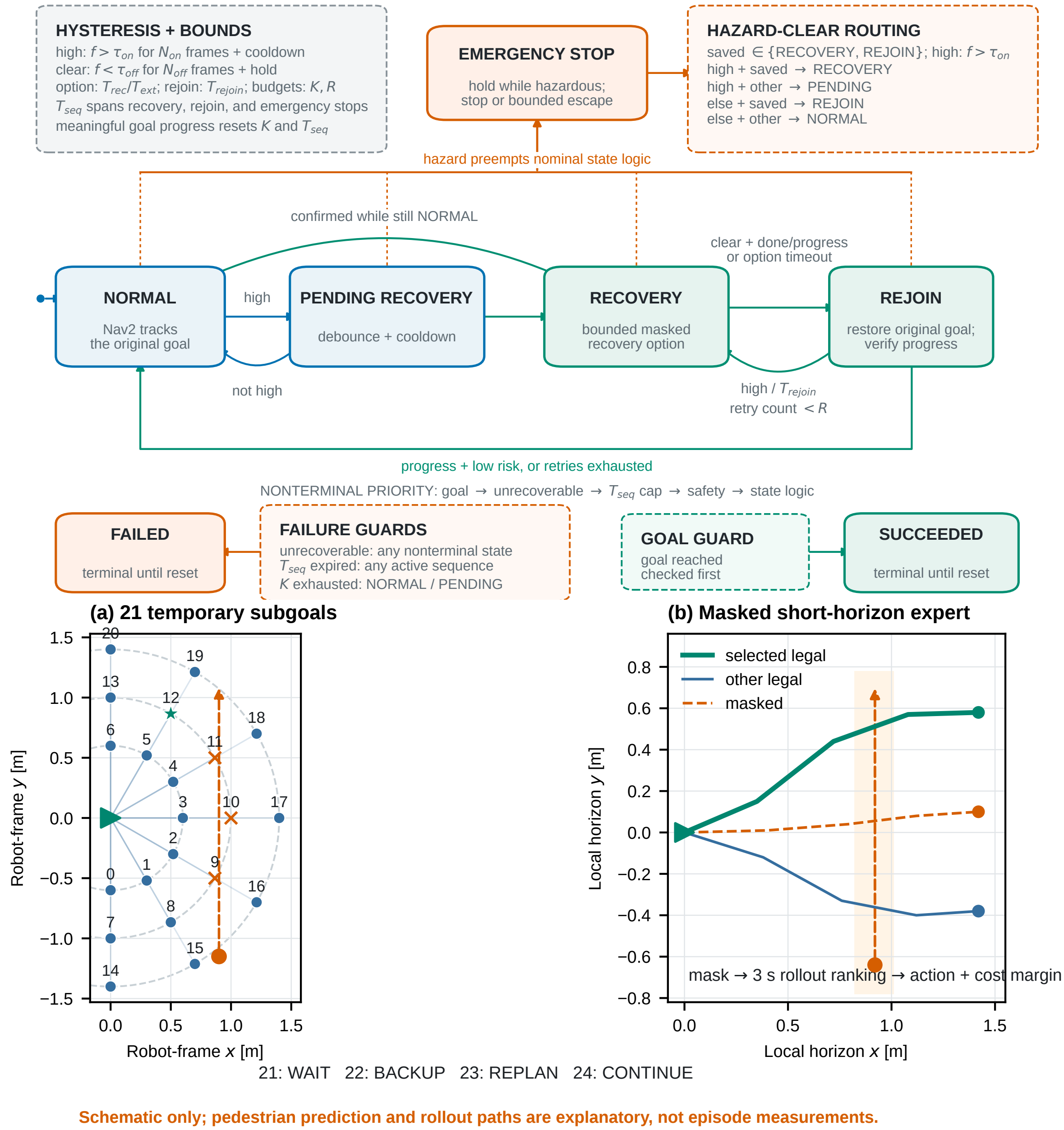

Fig. 2. Recovery design. Top: hysteretic, bounded state transitions with independent emergency priority. Bottom: schematic recovery actions and privileged expert ranking, not a recorded rollout. Mask-invalid candidates are never selectable, and pedestrian prediction is used only for training labels.

IV. Experimental Setup

A. Platform and Robot

Experiments execute in a pinned Ubuntu 22.04 Arena runtime with ROS2 Humble and Gazebo. Arena instantiates the world and Jackal skid-steer model; Nav2 NavFn plans the global path and DWB produces the nominal and temporary-goal velocity commands. All project nodes use simulation time. The map is 31.28×24.03 m; robot and pedestrian radii are 0.36 and 0.35 m. Jackal's planar LiDAR is resampled to 180 bins, and localized and physical goal tolerances are 0.25 and 0.30 m. Deployment receives only LiDAR, path, velocity, command, progress, planner-status, and rule histories. Simulator poses remain isolated for actor execution, expert labels, and evaluation.

LiDAR-visible cylindrical pedestrians follow deterministic one-shot routes at 5 Hz and 0.06–0.55 m/s. Swept-step guards hold motions inside a 0.90-m soft-yield region and prevent footprint crossing at 0.73 m. This produces paired, replayable interaction stress tests, not a model of human intent. The same actor guard is applied to every method and sees no method identity. It only modifies a pedestrian's proposed step: it neither overrides robot commands nor suppresses the logger's robot–pedestrian collision outcome, so a robot entering the physical

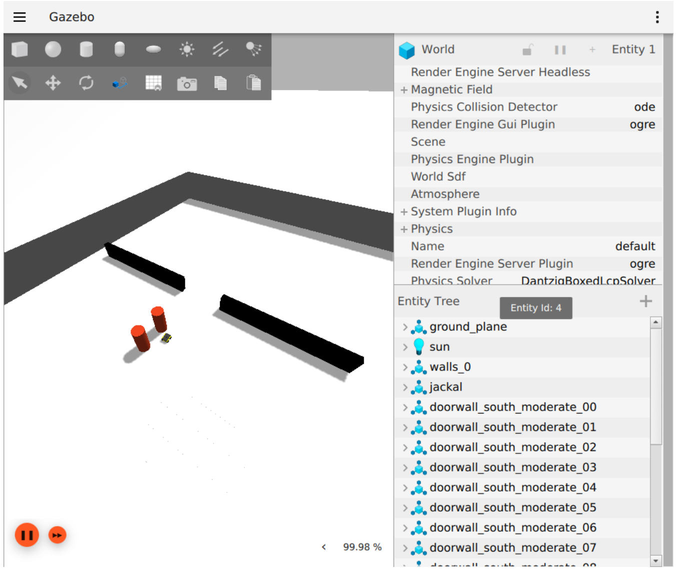

Fig. 3. Pixel capture from the mapped Gazebo GUI during execution of a frozen Arena doorway scene. The viewport shows the Jackal model, static doorway geometry, and LiDAR-visible cylindrical pedestrian proxies. This is a qualitative environment demonstration, not an onboard camera image or a numerical outcome; all comparisons come from paired raw logs and Parquet.

collision radius is still counted as a collision.

B. Scenarios, Splits, and Comparators

The benchmark covers eight interaction families: head-on corridor, doorway bottleneck, crossing flow, blind corner, group blocking, overtaking, opposite streams, and temporary blockage. Low, medium, and high density use one, two, and four pedestrians. All splits share the map and family templates, but use disjoint seeds, scenario IDs, and compiled physical realizations; therefore the test measures held-out interactions, not map or template generalization.

Validation has three repetitions in each of 24 family–density cells (72 conditions per method); held-out test has five (120 per method). Validation is used for code, configuration, and checkpoint selection. Test scenarios remain unexecuted until code, hashes, and checkpoints are frozen. Every method then receives identical starts, goals, pedestrian routes, seeds, planner profile, and 240-s limit. Artifact generation rejects missing conditions or mismatched pair metadata.

C. Calibration Audit and v6 Freeze

The first complete moderate-v5 Base validation retained 57 goal reaches and 15 collisions over all 72 declared conditions: 79.17% success exceeded the predeclared 75% maximum. The benchmark was therefore rejected without widening its threshold, and its test split was neither executed nor inspected. The validation diagnosis also found that Base reached all nine Crossing Flow conditions because the one-shot actors cleared the intersection before the robot arrived.

Moderate-v6 retains every v5 route, lane, static-clearance, collision-radius, actor-guard, and method parameter. Its only interaction change slows Crossing Flow from 0.35–0.55 to 0.18–0.28 m/s, aligning actor arrival with the validation-observed robot phase. New validation and test seed blocks begin at 77000 and 87000, respectively. Both splits were compiled and frozen before v6 simulation; at protocol freeze, the 120-condition test was still unexecuted and uninspected. No v6 test value enters this manuscript until the unchanged calibration gate passes and the complete artifact pipeline verifies all five methods on identical condition keys.

We compare four baselines with PGRR. *Base DWB* is NavFn/DWB with Arena's periodic-replanning behavior tree and no PGRR node. *Standard* changes only that tree to Arena's supplied replanning-and-recovery tree. *Heuristic* adds the observable detector, state machine, mask, and supervisor to Base, but selects legal actions deterministically. *Uniform BC* uses the same recovery hierarchy, network architecture, expert demonstrations, mask, and supervisor as PGRR, but omits DAgger aggregation and its train-only coverage shard. *PGRR* uses the frozen validation-selected DAgger checkpoint. Thus Heuristic isolates learned action selection and Uniform BC isolates aggregation; the expert is a training and diagnostic reference, not a deployment baseline. Margin weighting is retained as a negative offline result. PPO and a learned detector are not evaluated contributions.

D. Metrics and Statistics

Logs retain geometry, commands, planner and recovery state, rule scores, LiDAR, and paths. We report goal, collision, timeout, planner failure, SPL, path length, success-conditional time, minimum human distance, personal-space and discomfort measures, emergency stops, triggers, recovery success and duration, and intervention ratio. For episode i, $\mathrm{SPL}_i = S_i L_i / \max(L_i, P_i)$; success-conditioned time uses only $S_i = 1$. Personal-space ratio integrates $\mathbb{1}[d_h < 1.2\,\mathrm{m}]$ over observed time, while discomfort time uses $d_h < 1.0$ m. A trigger is an entry from NORMAL into recovery-active state, recovery success is a logged REJOIN→NORMAL transition, recovery duration integrates states PENDING through EMERGENCY-STOP, and intervention ratio additionally includes a non-CONTINUE action or command override. Emergency stops count entries, not samples. Full formulae and implementation pointers are in `docs/metrics.md`. Simulator/reset failures remain recorded and may be excluded only by the declared protocol; algorithm failures are never merged, removed, or retried.

Methods are paired on identical condition keys. The preregistered goal, collision, and timeout endpoints use exact McNemar tests and paired bootstrap intervals; planner failure remains descriptive. Continuous differences use two-sided Wilcoxon tests, paired intervals, and rank-biserial effects. A single Holm correction covers all predeclared comparator–endpoint tests. Confidence intervals use 10,000 resamples with seed 20260804; estimates, intervals, and effect sizes are reported regardless of significance.

E. Execution Integrity

Parallel workers use isolated ROS domains and simulator partitions; each run manifest records the exact worker count. A startup handshake holds actors until navigation, pose feedback, and logging are live. The runner refuses overwrite and records checkpoint and scenario hashes. A logical task permits two retries only after classified `SIMULATOR_FAILURE` or `INVALID_RESET`; collision, timeout, and planner failure are accepted outcomes. Complete five-method pairing is checked before statistics, tables, and figures are generated from Parquet, CSV, and JSON artifacts.

V. Results and Analysis

A. Closed-Loop Navigation Outcomes

Table I reports all retained terminal outcomes for the complete held-out three-density Gazebo benchmark. Of 600 logical method–episodes, 600 enter algorithm metrics and 0 are retained as simulator/reset exclusions. Base DWB reaches 70.8% of goals, collides in 29.2%, and times out in 0.0% of its valid episodes. PGRR reaches 90.8%, collides in 0.0%, and times out in 1.7%.

The table preserves every terminal category, so lower contact cannot be misread as successful recovery. Density- and family-resolved plots remain in the generated artifact bundle and accompanying technical report.

B. Paired Effects Against Four Baselines

On the 120 valid Base–PGRR pairs, the PGRR-minus-DWB goal-reaching difference is +20.0 [+12.5, +28.3] pp ($p_{\text{Holm}} < 0.001$), the collision difference is −29.2 [−37.5, −21.7] pp ($p_{\text{Holm}} < 0.001$), and the timeout difference is +1.7 [+0.0, +4.2] pp ($p_{\text{Holm}} = 1.000$). These endpoints are interpreted jointly: goal reaching measures task completion, while collision and timeout describe distinct failure modes.

Planner-failure rates are 0.0% for Base and 7.5% for PGRR (Δ =+7.5 pp). This is descriptive only: planner failure was not a preregistered inferential endpoint, so no post-hoc test is added.

For 83 jointly successful pairs, the PGRR-minus-DWB duration difference is +15.877 [+11.427, +20.834] s ($p_{\text{Holm}} < 0.001$) and the path difference is +1.886 [+1.205, +2.691] m ($p_{\text{Holm}} < 0.001$). These paired efficiency estimates condition on success and do not replace terminal outcomes.

Table II extends the same paired analysis to Standard, Heuristic, and Uniform BC and reports raw and globally Holm-adjusted tests, confidence intervals, and effect sizes without selecting a favorable scenario or seed after evaluation.

C. Recovery Behavior

The generated recovery audit retains triggers, successful REJOIN-to-NORMAL transitions, recovery time, intervention ratio, and emergency-stop entries for every method. The full table remains in the artifact bundle and technical report; these diagnostics do not replace terminal navigation outcomes.

D. Historical v1 Safety–Completion Boundary

The superseded v1 frozen test completed 64/64 logical episodes: Base and PGRR each covered 24 matched conditions, while Standard and Heuristic each covered eight high-density conditions. On the matched conditions, PGRR recorded 0/24 collisions versus Base's 19/24, but 16/24 timeouts versus 0/24; goal reaches were 8/24 versus 5/24, and the adjusted goal-reaching comparison was not significant (Holm $p = 0.75$). Thus v1 documents collision avoidance traded for completion, not general superiority. It is historical context only, is not a moderate-v6 result, and is excluded from every v6 estimate and hypothesis test.

E. Matched Execution Evidence

Aggregate tables are complemented by one matched Base–PGRR condition selected deterministically from the final `results.parquet`. The release renderer must join both method rows to their raw JSONL, metadata, and outcome sidecars, verify the scenario and commit hashes, and print each terminal outcome verbatim. This panel is illustrative ($n = 1$) and is never used to estimate an effect. The top-down trajectory and timeline are telemetry reconstructions, not camera screenshots; Figure 3 separately provides an actual Gazebo GUI pixel capture of the frozen environment.

F. Offline Imitation and Mask Diagnostics

The separate offline ablation in Table III compares Uniform BC, margin-weighted BC, and the selected DAgger checkpoint on the same scenario-disjoint validation states. Margin weighting is retained as a negative result rather than a selected contribution. The no-mask rows are counterfactual mask-dependence diagnostics and are not executed in simulation. They test expert agreement and invalid-action exposure, not a controlled closed-loop causal effect; the complete five-method closed-loop evaluation above remains the evidence for navigation performance.

VI. Limitations

The study uses planar LiDAR, a known two-dimensional map, and simulation localization that is more accurate than a deployed AMCL stack. Its deterministic pedestrians are useful for paired replay but do not capture human intent, group norms, perception uncertainty, or the diversity of real crowds. Minimum distance and discomfort time are therefore geometric proxies rather than validated human-comfort judgments. Their shared swept-step guard improves scenario repeatability but is itself a simplified response model; because it does not override robot control or collision labeling, it should not be read as a safety mechanism for any compared navigation method.

The detector, planning mask, learned policy, Nav2 controller, and emergency supervisor act jointly in the deployed hierarchy. Closed-loop outcomes do not attribute an effect to any one component, and the stopping supervisor provides no formal collision-avoidance guarantee. The privileged expert depends on simulator state and short-horizon motion prediction; its finite 25-action set cannot express arbitrary

TABLE I
HELD-OUT THREE-DENSITY GAZEBO OUTCOMES.

Method	n	Success [95% CI] ↑ (%)	Collision ↓ (%)	Timeout ↓ (%)	Planner fail. ↓ (%)	SPL ↑	Goal time ↓ (s)	$d_{\min}$ ↑ (m)
DWB	120	70.8 [62.2, 78.2]	29.2	0.0	0.0	0.703 ± 0.454	86.7 ± 7.1	0.89 ± 0.30
Standard	120	67.5 [58.7, 75.2]	32.5	0.0	0.0	0.671 ± 0.468	86.2 ± 7.2	0.91 ± 0.42
Heuristic	120	83.3 [75.7, 88.9]	0.8	5.0	10.8	0.795 ± 0.362	96.2 ± 16.3	0.93 ± 0.15
Uniform BC	120	86.7 [79.4, 91.6]	0.8	4.2	8.3	0.823 ± 0.330	100.5 ± 19.8	0.95 ± 0.15
PGRR	120	90.8 [84.3, 94.8]	0.0	1.7	7.5	0.831 ± 0.280	103.3 ± 21.1	0.95 ± 0.15

Note. All methods use identical scenario–density–seed conditions. Success intervals are two-sided 95% Wilson intervals. Collision, timeout, and planner failure are reported as distinct terminal outcomes and are not merged or omitted. SPL and $d_{\min}$ use all valid episodes; goal time is conditional on goal reaching. Continuous entries are mean ± sample SD. Simulator/reset rows are excluded from algorithm rates but retained in the artifacts (counts: DWB: 0; Standard: 0; Heuristic: 0; Uniform BC: 0; PGRR: 0). Bold identifies the proposed method.

TABLE II
PAIRED PGRR-MINUS-BASELINE TERMINAL-OUTCOME COMPARISONS ON THE HELD-OUT GAZEBO BENCHMARK.

Baseline	Endpoint	Test	Δ [95% CI]	p_{raw}	p_{Holm}	Effect	n_{pairs}
DWB	Goal reached	McNemar	+20.0 [+12.5, +28.3] pp	< 0.001	< 0.001	OR_H = 10.6	120
DWB	Collision	McNemar	−29.2 [−37.5, −21.7] pp	< 0.001	< 0.001	OR_H = 0.0141	120
DWB	Timeout	McNemar	+1.7 [+0.0, +4.2] pp	0.500	1.000	OR_H = 5	120
Standard	Goal reached	McNemar	+23.3 [+15.0, +31.7] pp	< 0.001	< 0.001	OR_H = 12.2	120
Standard	Collision	McNemar	−32.5 [−40.8, −24.2] pp	< 0.001	< 0.001	OR_H = 0.0127	120
Standard	Timeout	McNemar	+1.7 [+0.0, +4.2] pp	0.500	1.000	OR_H = 5	120
Heuristic	Goal reached	McNemar	+7.5 [+0.0, +15.8] pp	0.093	1.000	OR_H = 2.2	120
Heuristic	Collision	McNemar	−0.8 [−2.5, +0.0] pp	1.000	1.000	OR_H = 0.333	120
Heuristic	Timeout	McNemar	−3.3 [−7.5, +0.0] pp	0.219	1.000	OR_H = 0.273	120
Uniform BC	Goal reached	McNemar	+4.2 [−1.7, +10.8] pp	0.302	1.000	OR_H = 1.91	120
Uniform BC	Collision	McNemar	−0.8 [−2.5, +0.0] pp	1.000	1.000	OR_H = 0.333	120
Uniform BC	Timeout	McNemar	−2.5 [−6.7, +1.7] pp	0.453	1.000	OR_H = 0.455	120

Note. Differences are PGRR minus the named baseline. Binary-rate differences use percentage points (pp), episode-pair bootstrap 95% CIs, exact McNemar tests, and Haldane-corrected matched odds ratios OR_H. Continuous paired endpoints remain available in the technical report and machine-readable statistics. The reported Holm value belongs to one global family of 52 predeclared binary and continuous tests across all four baselines.

TABLE III
OFFLINE POLICY ABLATION ON 399 SCENARIO-DISJOINT VALIDATION STATES.

Model	Planning mask	Top-1 (%)	Top-3 (%)	Invalid (%)	Cost regret	Near-opt. (%)	Catastrophic (%)
Uniform BC	Enabled	88.5	95.7	0.0	0.271	88.7	0.0
Uniform BC	Disabled†	0.0	91.0	93.2	9.32×10^5	0.0	93.2
Margin-weighted BC	Enabled	88.5	95.7	0.0	0.271	88.7	0.0
Margin-weighted BC	Disabled†	0.0	91.7	93.2	9.32×10^5	0.0	93.2
Triggered DAgger	Enabled	92.2	98.5	0.0	0.031	93.2	0.0
Triggered DAgger	Disabled†	12.0	41.9	84.0	8.40×10^5	12.8	84.0

Note. Regret is measured in expert-cost units; its hard invalid-action penalty dominates mask-disabled rows. † Disabled-mask rows are counterfactual offline proposals and were never executed in closed loop. "Catastrophic" denotes selection of a hard-penalty action. Bold identifies the selected policy.

reciprocal negotiation. The expert is consequently a training and diagnostic reference, not a certified optimal controller.

All final conditions use one pinned simulator/runtime profile. We do not claim cross-simulator transfer, hardware validation, PPO refinement, or a learned failure detector. Split separation is over seeds and compiled physical realizations, not maps or interaction-family templates; five held-out repetitions per cell still cover a limited range of pedestrian behavior. Responsive human models, additional maps and planners, formal safety analysis, and real-robot trials are the most important extensions.

VII. CONCLUSION

We presented PGRR: Planning-Guided Failure-Triggered Recovery and Rejoin for Dynamic Social Navigation, an interface that retains DWB for nominal control and restricts learned intervention to interpretable temporary goals and recovery modes. A privileged rollout planner supplies supervision, DAgger covers learner-induced states, and planning masks constrain labeling and online inference. On the complete held-out three-density Gazebo benchmark, PGRR reaches 90.8% of goals versus 70.8% for DWB; the paired goal, collision, and timeout differences are +20.0 [+12.5, +28.3] pp, −29.2 [−37.5, −21.7] pp, and +1.7 [+0.0, +4.2] pp. Reporting those endpoints separately keeps task completion, contact, and conservative stagnation visible. Richer interaction prediction, additional runtime profiles, and hardware validation are the most direct next steps. The earlier v1 frozen test remains a boundary on interpretation: PGRR changed collisions from

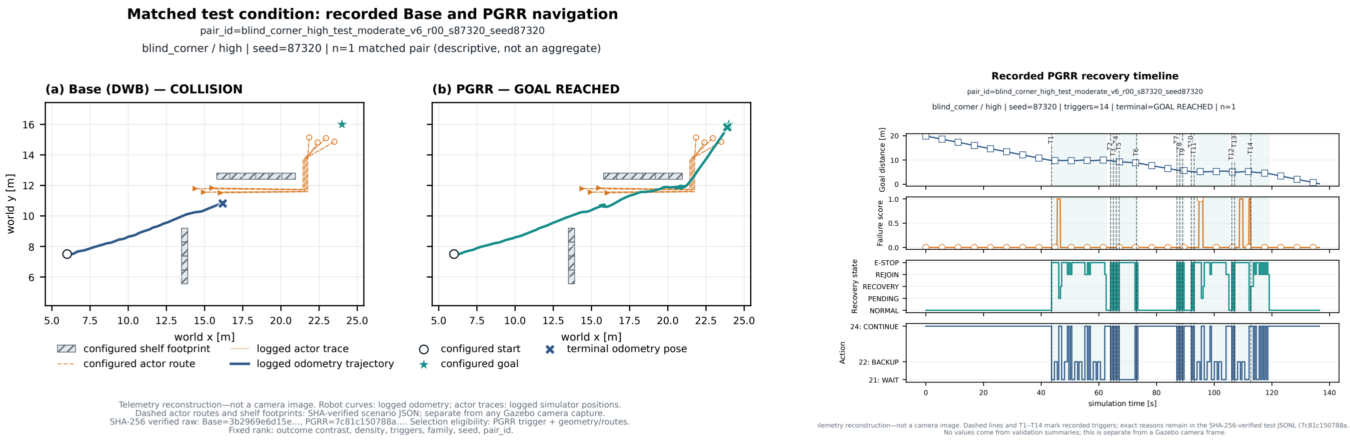

Fig. 4. Matched measured execution generated from final Parquet and raw episode streams. Left: Base and PGRR trajectories for the same condition, with the recorded terminal outcome printed for each method. Right: PGRR failure score, goal distance, recovery state, and action over simulation time. These are telemetry reconstructions, not simulator or onboard camera screenshots; the actual Gazebo scene is shown separately in Figure 3.

19/24 to 0/24 while timeouts changed from 0/24 to 16/24, and its 8/24 versus 5/24 goal-reaching difference was not significant after adjustment. Those historical v1 outcomes are not moderate-v6 evidence and are not pooled with the present benchmark.

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